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CUET UG

Biology

Human Health and Disease

Official Syllabus Unit VIII · Class 12 · CUET UG 2027 Cycle Notes

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Syllabus Alignment & Exam Blueprint

Sub-Topic	Official NTA CUET UG Syllabus Scope	High-Yield Exam Focus
Pathogens & Common Diseases	Common diseases in humans: Typhoid, pneumonia, common cold, amoebiasis, ascariasis, filariasis, ringworm.	Pathogen species, transmission vectors, target organs, diagnostic tests (Widal), symptoms, and life cycles.
Malarial Parasite Lifecycle	Life cycle of *Plasmodium* in human host and female *Anopheles* mosquito; symptoms and recurrence.	Sporozoites, liver schizogony, erythrocyte rupture, toxic **haemozoin**, gametocytes, gut fertilisation.
Innate Immunity	Innate immunity: Physical, physiological, cellular, and cytokine barrier systems.	Skin/mucus; stomach HCl/ lysozyme; PMNL-neutrophils/ macrophages; **interferons** against viral spread.
Acquired Immunity & Vaccines	Acquired immunity: Humoral and cell-mediated immunity; active and passive immunity; vaccination.	B & T cells; antibody H_2L_2 structure; IgG, IgA, IgM, IgE; graft rejection (CMI); colostrum & antivenom.
Allergy & Autoimmunity	Allergies, auto-immunity, and human immune system; primary and secondary lymphoid organs.	IgE & mast cell histamine release; loss of self-tolerance (rheumatoid arthritis); bone marrow, thymus, spleen, MALT.
AIDS & Cancer	AIDS (HIV structure, transmission, replication, prevention); Cancer (types, causes, detection, treatment).	Reverse transcriptase, CD4+ T-helper depletion, ELISA; loss of contact inhibition, metastasis, carcinogens.
Drugs & Alcohol Abuse	Drugs and alcohol abuse: Adolescence, addiction, dependence, effects, prevention, and control.	Opioids (poppy), cannabinoids (hemp), cocaine (coca); target receptors; tolerance; withdrawal syndrome.

CUET HIGH-YIELD EXAM ARCHITECTURE & WEIGHTAGE GUIDE

Human Health and Disease is one of the most consistently high-scoring units in CUET UG Biology, contributing **4 to 6 compulsory questions (16–24 marks)** per test session. Questions rigorously probe: (1) Match-the-following pairs connecting pathogens, vectors, and diagnostic tests; (2) The sequential stages of the *Plasmodium* life cycle across hosts; (3) Barrier classifications of innate immunity; (4) Antibody heavy/light chain formulas and functional roles (IgG, IgA, IgE); (5) HIV retroviral replication mechanisms; (6) Distinctions between benign and metastatic malignant tumors; and (7) Source plants and physiological receptors of psychoactive drugs.

Common Human Diseases & Pathogen Profiles

A disease represents a physiological or psychological departure from complete health. Infectious diseases are caused by **pathogens**—taxonomically diverse organisms including bacteria, viruses, fungi, protozoans, and helminths—that invade the human body, adapt to host microenvironments, and disrupt normal cellular metabolism.

Bacterial Diseases

- **Typhoid Fever:**
  - **Causative Organism:** *Salmonella typhi*, a pathogenic Gram-negative bacterium.
  - **Route of Transmission:** Enters the small intestine via contaminated food and water, migrating through blood to other organ systems.
  - **Clinical Manifestations:** Sustained high fever (39° C to 40° C), generalized weakness, abdominal pain, constipation, headache, and loss of appetite. In advanced untreated cases, intestinal perforation and death may ensue.
  - **Diagnostic Hallmark:** Confirmed serologically by the **Widal test** (detecting agglutinating antibodies against O and H antigens of *S. typhi*).
  - **Classic Case:** Mary Mallon ("Typhoid Mary"), an asymptomatic cook who unknowingly transmitted typhoid to dozens of families through prepared meals.
- **Pneumonia:**
  - **Causative Organisms:** *Streptococcus pneumoniae* (pneumococcus) and *Haemophilus influenzae*.
  - **Infection Site:** Infects the pulmonary **alveoli** (air-filled sacs). The alveoli become engorged with exudate fluid and inflammatory debris, severely impeding respiratory gas exchange.
  - **Transmission:** Inhaling respiratory droplets/aerosols expelled by infected individuals, or sharing utensils and glasses.
  - **Clinical Features:** Fever, chills, productive cough, and headache. In severe, decompensated episodes, the lips and fingernails turn **gray to bluish/cyanotic** due to hypoxemia.

Viral Diseases

- **Common Cold:**
  - **Causative Agent: Rhinoviruses**, representing a diverse family of non-enveloped RNA viruses.
  - **Anatomical Target:** Infects the mucosal lining of the **nose and upper respiratory passage**, but critically **does NOT infect the lungs** (contrasting fundamentally with pneumonia).
  - **Transmission:** Inhaling infectious aerosols from coughs/sneezes, or indirect contact through contaminated fomites (pens, books, door handles, computer keyboards).
  - **Symptoms & Duration:** Nasal congestion, profuse watery discharge, sore throat, hoarseness, cough, headache, and malaise. Self-limiting, resolving within **3 to 7 days**. Antibiotics have zero therapeutic efficacy against rhinoviruses.

Protozoan Diseases

- **Amoebiasis (Amoebic Dysentery):**
  - **Causative Organism:** *Entamoeba histolytica*, a microscopic protozoan parasite inhabiting the human **large intestine**.
  - **Mechanical Vector:** The common **housefly** (*Musca domestica*) acts as a passive mechanical carrier, transferring cysts from human fecal deposits to unshielded food and water supplies.
  - **Pathology & Symptoms:** Constipation alternating with loose stools, severe abdominal cramps, and stools containing **excess mucus and blood clots**. Ulceration of mucosal walls can occur.
- **Malaria:**
  - **Causative Agents:** Microscopic protozoans of the genus *Plasmodium*:
    - *Plasmodium vivax* (benign tertian malaria; 48-hour cycle).
    - *Plasmodium malariae* (quartan malaria; 72-hour cycle).

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- *Plasmodium falciparum* (malignant tertian malaria; most lethal, causing cerebral complications, severe anemia, and renal failure).

Helminthic & Fungal Diseases

- **Ascariasis:**
  - **Pathogen:** *Ascaris lumbricoides*, a common intestinal roundworm endoparasite.
  - **Transmission:** Ingestion of embryonated eggs present in soil, water, and unwashed vegetables contaminated with human feces.
  - **Symptoms:** Internal hemorrhage, muscular aches, fever, anemia, and physical obstruction of the intestinal lumen under heavy worm burdens.
- **Filariasis (Elephantiasis):**
  - **Pathogen:** Filarial nematodes, primarily *Wuchereria bancrofti* and *Wuchereria malayi*.
  - **Transmission Vector:** Bite of an infected **female Culex mosquito**.
  - **Pathogenesis:** Adult worms reside inside the **lymphatic vessels** of the lower limbs and genital tract for years, triggering chronic inflammatory fibrosis, lymphatic obstruction, and gross enlargement of limbs and scrotum (**elephantiasis**).
- **Ringworm:**
  - **Causative Fungi:** Genera *Microsporum*, *Trichophyton*, and *Epidermophyton*.
  - **Transmission:** Direct skin contact, or fomites such as shared towels, clothing, and combs.
  - **Pathology:** Dry, scaly, erythematous, intensely pruritic (itchy) ring-like epidermal lesions. Fungal proliferation is strongly accelerated by warmth and perspiration in skin folds (groin, between toes).

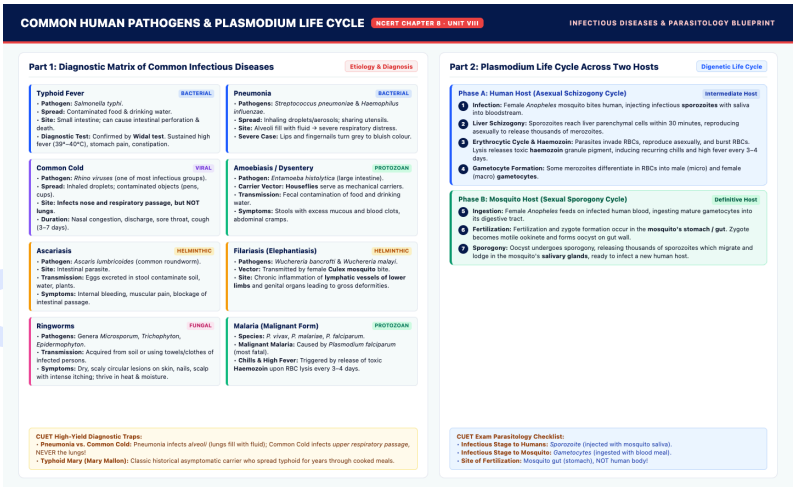


Figure 1: Malaria life cycle & human pathogen matrix — tracing Plasmodium development across the human host and female Anopheles vector, alongside diagnostic signatures, vectors, and clinical hallmarks of 8 major human infectious diseases.

The Malarial Parasite Life Cycle

Plasmodium requires **two hosts** to complete its biological life cycle (digenetic):

1. **Human Host:** Primary site of asexual reproduction (schizogony).
2. **Female Anopheles Mosquito Vector:** Definitive host where sexual reproduction (gametocyte fusion and sporogony) occurs.

Sequential Stages in the Human Host

1. **Inoculation:** An infected female *Anopheles* mosquito bites a human, injecting motile infectious **sporozoites** present in its salivary secretions into the bloodstream.
2. **Hepatic Phase (Exo-erythrocytic Schizogony):** Sporozoites rapidly migrate to the liver within 30 minutes, penetrating parenchymal hepatocytes where they multiply asexually, ultimately rupturing the liver cells to liberate thousands of **merozoites**.

3. **Erythrocytic Phase (Intra-erythrocytic Schizogony):** Merozoites penetrate red blood cells (erythrocytes), forming a feeding trophozoite (signet-ring stage), followed by an erythrocytic schizont.
4. **Haemozoin Release & Paroxysm:** The RBC ruptures, releasing new merozoites alongside a toxic, crystalline, dark-brown pigment called **haemozoin** (derived from the metabolic breakdown of host hemoglobin). Haemozoin triggers the synchronous release of pyrogens, inducing the characteristic **chills and high recurring fever** every 3 to 4 days.
5. **Gametocytogenesis:** Within circulating RBCs, some merozoites differentiate into sexual stages: male microgametocytes and female macrogametocytes.

Sequential Stages in the Mosquito Vector

1. **Gametocyte Ingestion:** When a female *Anopheles* mosquito takes a blood meal from an infected human, it ingests both micro- and macrogametocytes.
2. **Fertilisation:** Within the mosquito's midgut lumen, the gametocytes emerge and undergo fertilisation, producing a diploid motile zygote called an **ookinete**.
3. **Oocyst Formation:** The ookinete penetrates the peritrophic matrix and gut wall, encysting on the outer stomach wall as an **oocyst**.
4. **Sporogony:** Inside the oocyst, repeated nuclear divisions generate thousands of slender, spindle-shaped **sporozoites**.
5. **Salivary Migration:** The mature oocyst ruptures, releasing sporozoites into the mosquito's hemolymph, whence they migrate to and pool inside the **salivary glands**, ready to infect a new human host upon the next bite.

Innate Immunity — The Inborn Non-Specific Defences

Immunity is the host's overall physiological capacity to resist and combat foreign pathogens. It is partitioned into two distinct arms: **innate (non-specific)** and **acquired (adaptive/specific)** immunity.

Innate immunity is active from birth, confers immediate broad-spectrum protection, and does not possess immunological memory. It comprises four functional barrier systems:

Barrier Category	Mode of Action	Physiological Anatomical Examples
Physical Barriers	Impenetrable mechanical shields preventing microbial entry into body tissues.	Keratinized epidermis of the **skin**; **mucus coating** of the respiratory, gastrointestinal, and urogenital epithelial linings that traps microbes.
Physiological Barriers	Chemical microenvironments and secretions that inhibit microbial growth or destroy cell walls.	**Hydrochloric acid (HCl)** in gastric juice (pH 1.5–2.0); **lysozyme** (cleaves peptidoglycan) in saliva and tears; bile salts.
Cellular Barriers	Specialized phagocytic leukocytes that engulf, internalize, and enzymatically digest pathogens.	**PMNL-neutrophils** (polymorphonuclear leukocytes); circulating **monocytes**; tissue-resident **macrophages**; **Natural Killer (NK) cells** (cytolytic).
Cytokine Barriers	Soluble signaling glycoproteins secreted by infected cells to confer non-specific antiviral resistance.	**Interferons (IFN-α, IFN-β)** secreted by virus-infected cells, stimulating uninfected neighbor cells to synthesize antiviral translation-inhibitory proteins.

Acquired Immunity, Antibodies & Vaccination

Acquired (adaptive) immunity is pathogen-specific, develops throughout an individual's lifetime following antigen exposure, and is characterized by **immunological memory**.

Functional Architecture of Acquired Immunity

- **Antigen Specificity:** Capacity to discriminate among different microbial epitopes with high fidelity.
- **Immunological Memory:**
  - **Primary Response:** When the immune system encounters an antigen for the first time, response generation is slow and of low intensity. Memory cells are generated.
  - **Secondary (Anamnestic) Response:** Subsequent re-exposure to the identical antigen elicits a rapid, heightened, high-affinity response because memory B and T cells immediately recognize the antigen.

The Antibody Molecule (H_2L_2)

Antibodies (immunoglobulins, Ig) are complex glycoproteins synthesized by differentiated plasma cells:

- **Chain Composition:** Each antibody monomer possesses **4 polypeptide chains**:
  - **Two identical Heavy (H) chains** (50 kDa each).
  - **Two identical Light (L) chains** (25 kDa each).
  - The molecular formula is designated $\boldsymbol{H_2L_2}$.
- **Disulfide Linkages:** Heavy and light chains are joined by covalent **interchain disulfide bonds** ($-S-S-$).
- **Variable & Constant Regions:**
  - **Fab region (Fragment antigen-binding):** Formed by the variable domains (V_H and V_L), creating the hypervariable **antigen-binding site** (paratope) at the tips of the Y-fork. Each monomer has two identical antigen-binding sites (bivalent).
  - **Fc region (Fragment crystallizable):** Formed by constant domains (C_H), mediating effector functions (macrophage binding, complement activation, placental transfer).

Classes of Immunoglobulins

Class	Structural Form	High-Yield Physiological Role
IgG	Monomer (80% serum Ig)	Most abundant circulating antibody; uniquely crosses the human **placenta** to confer passive immunity to the developing fetus.
IgA	Dimer with secretory piece	Abundant in seromucous secretions (tears, saliva, colostrum); guards mucosal surfaces against microbial adherence.
IgM	Pentamer with J-chain	First class synthesized during primary immune response; potent agglutinator with 10 antigen-binding sites.
IgE	Monomer	Binds high-affinity Fc receptors on **mast cells and basophils**; triggers release of histamine in **allergic hypersensitivity** and fights helminthic parasites.
IgD	Monomer	Acts as an antigen receptor on mature naive B-lymphocyte surfaces; regulates B-cell activation.

Two Arms: Humoral vs Cell-Mediated Immunity (CMI)

- **Humoral Immunity:** Mediated by antibodies produced by **B-lymphocytes** that circulate freely in extracellular fluids (blood plasma and lymph). Highly effective against extracellular bacteria, free viruses, and soluble toxins.
- **Cell-Mediated Immunity (CMI):** Mediated directly by **T-lymphocytes** (Cytotoxic CD8+ T cells and Helper CD4+ T cells). It does not involve circulating antibodies. CMI is crucial for destroying intracellular pathogens (viruses inside host cells, *Mycobacterium tuberculosis*), eliminating cancer cells, and is primarily responsible for **graft rejection** during organ transplantation. Tissue typing and HLA matching are

mandatory before transplants, and recipients require lifelong immunosuppressants (such as Cyclosporin A).

Active vs Passive Immunity

Parameter	Active Immunity	Passive Immunity
Mechanism	Host's **own cells** actively produce antibodies upon antigen exposure.	Pre-formed antibodies are **transferred ready-made** into the recipient.
Onset	Slow onset; takes days to weeks to generate protective titers.	Immediate, rapid protection.
Memory	Generates long-lived memory B and T cells; protection lasts years/lifetime.	Zero memory cells produced; protection wanes as foreign antibodies degrade.
Examples	Natural infection; active vaccination with attenuated or killed microbes.	**Colostrum** (rich in maternal IgA); transplacental IgG; anti-tetanus serum (ATS); anti-snake venom.

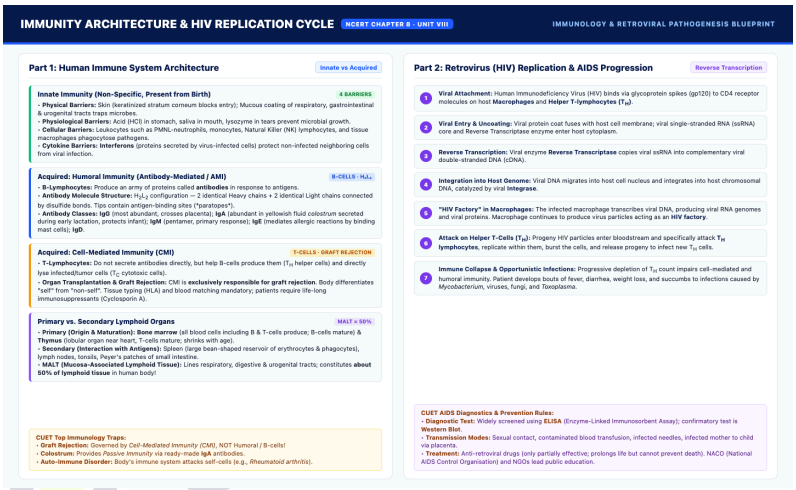


Figure 2: Immunity architecture & HIV replication cycle — structural mechanics of antibody molecules (H2L2), humoral vs cell-mediated pathways, and the retroviral lifecycle from CD4 receptor engagement to viral budding and immune depletion.

Immune System Anatomy & Immunopathology

Lymphoid Organs: Primary vs Secondary

Lymphoid organs are anatomical sites where lymphocytes originate, differentiate, mature, and interact with foreign antigens.

- **Primary Lymphoid Organs:**
  - **Sites: Bone marrow** and **Thymus**.
  - **Function:** Immature lymphocytes acquire antigen receptors and undergo maturation into antigen-sensitive cells without foreign antigen stimulation.
  - **Bone Marrow:** Main lymphoid tissue where all blood cells (erythrocytes, leukocytes, platelets) and B/T progenitors originate. B-lymphocytes undergo complete maturation here.
  - **Thymus:** Lobed organ situated near the heart, beneath the breastbone (sternum). It is large at birth but steadily atrophies and shrinks with advancing age. T-lymphocytes migrate from bone marrow to the thymus to undergo maturation and tolerance education.
- **Secondary Lymphoid Organs:**
  - **Sites: Spleen**, **lymph nodes**, **tonsils**, **Peyer's patches** of the small intestine, and the **appendix**.
  - **Function:** Provide sites where mature immunocompetent lymphocytes encounter antigens, undergo clonal expansion, and differentiate into effector cells.

- **Spleen:** Large bean-shaped organ containing lymphocytes and phagocytes. Acts as a physiological blood filter trapping blood-borne microorganisms and worn-out erythrocytes (the **"graveyard of RBCs"** and reservoir of erythrocytes).
- **Lymph Nodes:** Small nodular structures located along lymphatic channels. They filter lymph fluid, trapping foreign particulate antigens that activate specific lymphocytes.
- **MALT (Mucosa-Associated Lymphoid Tissue):** Lymphoid tissue embedded within the mucosal lining of major physiological tracts (respiratory, digestive, urogenital). MALT accounts for **approximately 50%** of all lymphoid tissue in the human body.

Allergies (Hypersensitivity)

- **Definition:** An exaggerated, inappropriate immune inflammatory response to normally innocuous environmental substances (allergens: pollen grains, dust mites, animal dander, fungal spores).
- **Mechanism:** Mediated primarily by **IgE antibodies**. Upon initial exposure (sensitisation), IgE antibodies attach to high-affinity Fc receptors on the surface of **mast cells** and basophils.
- **Degranulation:** Upon re-exposure, allergen cross-links surface-bound IgE, triggering rapid degranulation and explosive release of inflammatory mediators: **histamine** and **serotonin**.
- **Clinical Features:** Sneezing, watery eyes, running nose, bronchoconstriction, and urticaria.
- **Therapy:** Antihistamines, adrenaline, and corticosteroids rapidly alleviate acute allergic symptoms.

Autoimmunity

Under normal physiological conditions, the adaptive immune system displays **self-tolerance**—the capacity to distinguish foreign non-self molecules from self-antigens. When this tolerance breaks down, the immune system mounts a humoral or cell-mediated attack against the host's own healthy cells and tissues, causing **auto-immune disease**.

- **Archetype: Rheumatoid arthritis** (chronic progressive autoimmune inflammation of the synovial membranes of joints). Other examples include Hashimoto's thyroiditis and Myasthenia gravis.

AIDS — Acquired Immunodeficiency Syndrome

AIDS represents the terminal clinical manifestation of chronic infection by the **Human Immunodeficiency Virus (HIV)**, characterized by profound immunosuppression and susceptibility to opportunistic infections.

Molecular Structure of HIV

- **Classification:** HIV is an enveloped animal virus belonging to the **Retrovirus** family (genus *Lentivirus*).
- **Genome:** Comprises **two identical molecules of single-stranded RNA** (ssRNA diploid genome).
- **Enzymes:** Contains essential viral enzymes: **Reverse transcriptase**, **Integrase**, and **Protease**.
- **Envelope:** Phospholipid bilayer derived from the host membrane, embedded with viral glycoprotein spikes (**gp120** docking glycoprotein and **gp41** transmembrane stalk).

Pathogenesis & Replication Cycle

1. **Attachment & Fusion:** Viral gp120 binds specifically to the **CD4 receptor** present on the surface of helper T-lymphocytes (T_H cells) and monocytes/macrophages, alongside chemokine coreceptors (CCR5/CXCR4). The viral envelope fuses with the host plasma membrane, injecting the viral capsid core into the cytoplasm.
2. **Reverse Transcription:** Viral reverse transcriptase transcribes the viral ssRNA genome into complementary single-stranded DNA, and subsequently into double-stranded viral DNA (RNA → DNA).
3. **Integration:** The viral dsDNA enters the host nucleus and is integrated into the host genome by the viral enzyme **integrase**, establishing a provirus state.
4. **The "HIV Factory":** In infected **macrophages**, the virus continuously directs transcription of new viral RNAs and proteins without immediately killing the macrophage. Macrophages act as persistent reservoirs and manufacturing hubs (**"HIV factory"**).
5. **CD4+ T-Helper Cell Depletion:** HIV attacks and replicates inside **helper T-lymphocytes (T_H cells)**. Progeny virions bud from the membrane, lysing and progressively depleting the total CD4+ count.

6. **Opportunistic Infections:** As CD4+ counts fall below critical thresholds (under 200 cells/μL), the patient suffers from bouts of fever, diarrhea, and weight loss, becoming prey to lethal **opportunistic infections** caused by *Mycobacterium*, *Pneumocystis jirovecii* (fungal pneumonia), cytomegalovirus, and *Toxoplasma*.

Modes of Transmission & Clinical Testing

- **Transmission Routes:** Sexual contact with infected partners; transfusion of contaminated blood/blood products; sharing contaminated hypodermic needles (intravenous drug users); vertical transmission from infected mother to child through placenta or breast milk.
- **Non-Transmission:** HIV is **NOT transmitted** through casual social contact (hugging, shaking hands, sharing utensils), respiratory droplets, saliva, tears, or mosquito/insect bites.
- **Diagnosis:**
  - **Screening Test: ELISA** (Enzyme-Linked Immunosorbent Assay).
  - **Confirmatory Test: Western Blotting** (detects antibodies specific to viral proteins gp120, gp41, p24).
  - **Quantitative Viral Load:** Reverse Transcriptase-Polymerase Chain Reaction (RT-PCR).
- **Treatment:** Anti-Retroviral Therapy (ART)—using reverse transcriptase inhibitors and protease inhibitors—suppresses viral replication and prolongs survival, but cannot eradicate the provirus.

Cancer Biology & Carcinogenesis

Cancer represents a breakdown of the regulatory mechanisms that govern normal cell growth, division, and spatial tissue architecture.

Cellular Properties of Cancer Cells

- **Loss of Contact Inhibition:** Normal mammalian cells exhibit **contact inhibition**—a regulatory phenomenon wherein physical contact with neighboring cells inhibits further cell division. Cancer cells completely lose contact inhibition, continuing uncontrolled mitosis to form disorganized cellular masses termed **tumors (neoplasms)**.
- **Immortality:** Activation of telomerase enzyme allows indefinite replication cycles without entering senescence.

Types of Tumors

Feature	Benign Tumor	Malignant Tumor (True Cancer)
Growth & Boundary	Remains strictly localized to its tissue of origin; encapsulated by fibrous connective tissue.	Unencapsulated; invades and destroys surrounding healthy anatomical tissues.
Growth Rate	Slow, controlled enlargement.	Rapid proliferation, competing with normal cells for vital nutrients.
Metastasis	**Completely absent.** Never spreads to distant anatomical sites.	**Present.** Cells detach, enter blood/lymph, lodge in distant organs to form secondary tumors.
Clinical Danger	Causes minimal damage unless physically pressing on vital structures.	Extremely fatal without aggressive oncological intervention.

- **Metastasis:** The detachment of malignant cells from the primary tumor and their dissemination via blood or lymphatic circulation to seed secondary metastatic colonies in distant organs. It is the **most dreaded diagnostic hallmark** of malignant neoplasms.

Carcinogens & Oncogenic Transformation

Cancer-inducing agents are known as **carcinogens**:

- **Physical Carcinogens:** Ionizing radiation (X-rays, gamma rays) and non-ionizing radiation (UV rays) induce DNA double-strand breaks, thymine dimers, and mutational cascades.
- **Chemical Carcinogens:** Tobacco smoke (contains polycyclic aromatic hydrocarbons like benzopyrene, strongly linked to lung cancer), coal tar, arsenic, and aniline dyes.
- **Biological Carcinogens:**
  - **Oncogenic Viruses:** Viruses carrying viral oncogenes (*v-onc*) that integrate into host chromosomes.
  - **Cellular Oncogenes:** Normal mammalian cells harbor genes called **proto-oncogenes (cellular oncogenes, *c-onc*)** that regulate normal growth. When mutated or inappropriately activated by carcinogens, they convert into active **oncogenes**, driving neoplastic transformation.

Diagnostic Modalities & Therapeutic Interventions

- **Diagnostic Methods:**
  - **Biopsy & Histopathology:** A suspected tissue fragment is excised, sectioned, stained, and examined under a microscope for cellular atypia and mitotic abnormalities.
  - **Imaging Techniques:** Radiography (X-rays), Computed Tomography (**CT**—provides 3D anatomical images using X-rays), and Magnetic Resonance Imaging (**MRI**—uses intense magnetic fields and non-ionizing radiation to safely detect pathological tissue changes).
  - **Molecular & Immunological Tests:** Monoclonal antibodies against cancer-specific antigens; DNA probes detecting inherited susceptibility oncogenes (e.g., *BRCA1/2* for breast cancer).
- **Therapeutic Approaches:**
  - **Surgical Excision:** Physical removal of localized tumors.
  - **Radiation Therapy:** Tumor cells are targeted with lethal doses of ionizing radiation while shielding surrounding normal tissue.
  - **Chemotherapy:** Administration of antineoplastic pharmacological agents (e.g., vincristine, vinblastine, cisplatin) that inhibit cellular division. Side effects include alopecia (hair loss) and anemia.
  - **Immunotherapy:** Tumors frequently evade immune detection. Administering **biological response modifiers** such as **α-interferon** stimulates the patient's own cytotoxic immune system to destroy tumor cells.

Drugs, Tobacco and Alcohol Abuse

Adolescence, Addiction and Dependence

- **Adolescence:** The transitional developmental period between 12 and 18 years of age linking childhood with biological adulthood. High curiosity, vulnerability, peer pressure, and risk-taking behavior render adolescents prone to substance abuse.
- **Addiction:** A psychological attachment to certain pleasurable sensations (euphoria, temporary well-being) associated with drug or alcohol consumption.
- **Tolerance:** With chronic repeated usage, cell surface receptors tolerate the substance, necessitating progressively **higher doses** to achieve the desired psychological effect.
- **Dependence & Withdrawal Syndrome:** The physical adaptation of the body to the substance. If drug or alcohol consumption is abruptly ceased, the patient exhibits a distressing **withdrawal syndrome** characterized by intense anxiety, tremors, sweating, nausea, and vomiting.

Classification of Commonly Abused Drugs

Drug Category	Plant Source & Active Principle	Pharmacological Mechanism & Effects
Opioids	Obtained from the latex of the opium poppy ***Papaver somniferum***. Active principle: **Morphine**. Acetylation of morphine yields **Heroin** (diacetylmorphine, "smack").	Bind specific **opioid receptors** located in the Central Nervous System (CNS) and Gastrointestinal tract. Morphine is a potent clinical analgesic and sedative. Heroin is a powerful **depressant** that slows body functions.

Drug Category	Plant Source & Active Principle	Pharmacological Mechanism & Effects
Cannabinoids	Derived from the inflorescences of the Indian hemp plant ***Cannabis sativa***. Formulations include marijuana, hashish, charas, and ganja.	Interact with specific **cannabinoid receptors** predominantly located in the brain. They impair cognitive function and exert marked effects on the **cardiovascular system**. Highly abused by certain athletes.
Cocaine (Coca alkaloid)	Extracted from the leaves of the South American coca shrub ***Erythroxylum coca***. Commonly called "coke" or "crack".	Interferes with the reuptake of the neurotransmitter **dopamine**. Acts as a potent **CNS stimulant**, producing euphoria and surges of energy. Excessive dosages precipitate auditory and visual **hallucinations**.
Hallucinogens	From plants like ***Atropa belladonna*** and ***Datura***, as well as synthetic lysergic acid diethylamide (LSD).	Alter sensory perceptions, induce profound distortions of reality, time, and space, and precipitate severe psychological psychosis.

- **Tobacco & Nicotine:** Tobacco contains the toxic alkaloid **nicotine**, which stimulates the adrenal medulla to release adrenaline and nor-adrenaline into the blood, acutely elevating arterial blood pressure and heart rate. Smoking causes carbon monoxide (CO) to bind hemoglobin, forming carboxyhemoglobin and reducing oxygen-carrying capacity. It dramatically elevates incidences of lung, bladder, and throat cancer, bronchitis, and emphysema.
- **Alcohol:** Chronic ethanol ingestion induces peripheral neuropathy, cardiomyopathy, and extensive **liver cirrhosis** (fibrotic scarring replacing functional hepatocytes).

Key Facts & Terms Cheatsheet

MASTER DIAGNOSTIC & CONCEPTUAL CHEATSHEET

Key Subject Pointer	Core NCERT Mechanism / Definition	Exam Identifier
Typhoid Diagnosis	Serological agglutination detecting antibodies against *Salmonella typhi*.	Widal Test
Pneumonia Pathology	Fluid-filled alveoli hindering gas exchange; cyanotic blue lips/nails.	*S. pneumoniae*
Common Cold	Rhinoviruses infecting nose and upper airway; strictly spares the lungs.	3–7 Days duration
Toxic Malarial Agent	Insoluble crystalline pigment released when erythrocytes rupture.	Haemozoin
Infectious Malarial Stage	Spindle-shaped motile forms injected into human blood by mosquito.	Sporozoites
Filariasis Pathology	Nematodes residing in lower limb lymphatic vessels; elephantiasis.	*Wuchereria bancrofti*
Cytokine Barrier	Antiviral glycoproteins secreted by virus-infected cells.	Interferons
Placental Antibody	Smallest, most abundant immunoglobulin that traverses the placenta.	IgG
Colostrum Antibody	Secretory dimer in yellowish early breast milk conferring passive immunity.	IgA
Allergic Antibody	Binds mast cells, triggering explosive release of histamine and serotonin.	IgE

Key Subject Pointer	Core NCERT Mechanism / Definition	Exam Identifier
Organ Graft Rejection	Immune arm responsible for recognizing non-self MHC on foreign tissue.	Cell-Mediated (CMI)
Primary Lymphoid Organs	Sites where immature lymphocytes differentiate and mature.	Bone marrow & Thymus
Mucosal Lymphoid Tissue	Accounts for roughly 50 percent of all lymphoid tissue in humans.	MALT
AIDS Virus	Enveloped retrovirus with two single-stranded RNA molecules and reverse transcriptase.	HIV (CD4+ target)
AIDS Diagnostic Sequence	Enzyme-linked immunosorbent assay followed by Western blot confirmation.	ELISA → Western Blot
Cancer Spread	Dissemination of malignant cells to distant sites via circulation.	Metastasis
Biological Response Modifier	Glycoprotein activating immune destruction of neoplastic cells.	α-Interferon
Heroin Synthesis	Diacetylmorphine produced by acetylation of natural morphine.	*Papaver somniferum*
Cocaine Mechanism	Interferes with synaptic dopamine reuptake, driving CNS stimulation.	*Erythroxylum coca*

Common Errors and Special Cases

HIGH-YIELD CUET PITFALLS & DIAGNOSTIC DISTINCTIONS

Common Mistake	Scientifically Verified Reality	The Exact Trap Mechanism
Confusing Pneumonia with Common Cold	Pneumonia infects the **alveoli of the lungs**; the common cold affects **only the nose and respiratory passage**, strictly sparing the lungs.	NTA questions test whether fluid accumulation occurs in alveoli or purely in upper nasopharyngeal membranes.
Swapping Malaria and Filariasis Vectors	Malaria is transmitted by **female Anopheles**; Filariasis is transmitted by **female Culex**; Dengue/ Chikungunya by **female Aedes**.	Exam questions frequently create vector-disease mismatch matrices; never confuse *Culex* with *Anopheles*.
Classifying Colostrum as Active Immunity	Colostrum provides **passive immunity** because ready-made maternal IgA is received without host antigen challenge.	Any ready-made antibody transfer (colostrum, antivenom, ATS) is passive; memory cells are never formed.
Attributing Graft Rejection to Antibodies	Graft rejection is mediated strictly by **Cell-Mediated Immunity (CMI)** via T-lymphocytes, NOT humoral antibodies.	Organ transplant rejection questions test the direct role of cytotoxic T cells and tissue matching.
Believing Benign Tumors Metastasise	Only **malignant neoplasms** undergo metastasis; benign tumors remain localized and encapsulated.	Metastasis is the single defining biological property that separates malignant from benign growths.

Common Mistake	Scientifically Verified Reality	The Exact Trap Mechanism
Confusing Opioids with Cannabinoids	Opioids are **depressants** (slow body functions); Cannabinoids act on cardiovascular/brain receptors; Cocaine is a **stimulant**.	Remember the three operational verbs: opioids **depress**, cocaine **stimulates**, cannabinoids **distort**.

The Whole Chapter on One Page

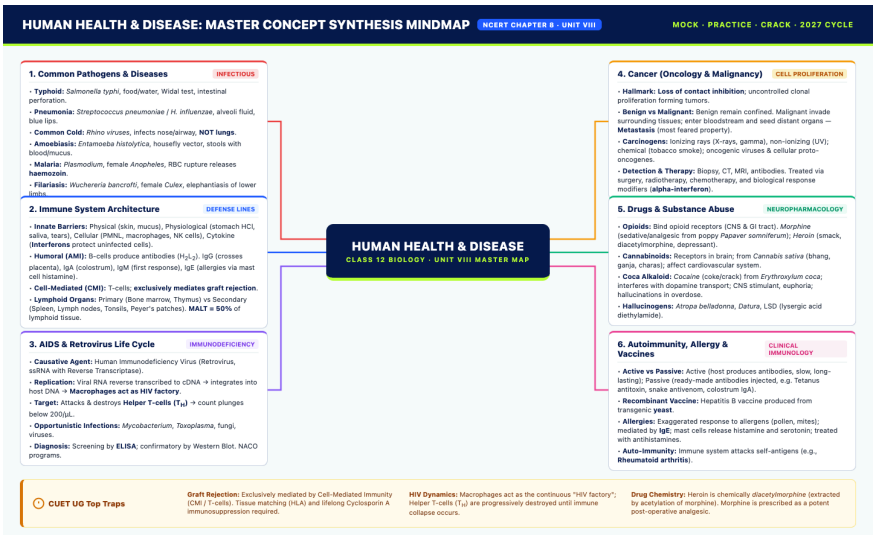


Figure 3: Master synthesis mindmap — high-yield integrated architecture of Human Health & Disease covering pathogen profiles, immune defenses, AIDS, oncology, and pharmacology.

Practice Questions — Part 1 (Q1 to Q5)

Q1. A patient presents with sustained high fever (39°C to 40°C), severe headache, weakness, stomach pain, and constipation. Laboratory evaluation confirms a positive Widal test. Which of the following statements regarding the causative pathogen is correct?

(A) It is a virus that infects the alveoli of the lungs causing fluid congestion
(B) It is the Gram-negative bacterium *Salmonella typhi* that enters through contaminated food and water
(C) It is an amoebic protozoan that establishes residence exclusively in the lumen of the large intestine
(D) It is a filarial nematode transmitted by the bite of an infected female *Culex* mosquito

Sol. Q1. Typhoid fever is caused by the pathogenic bacterium *Salmonella typhi*. It enters the human body via contaminated food and drinking water, reaches the small intestine, and migrates to other organs through the bloodstream. Classical symptoms include sustained fever (39° C $- 40^{\circ}$ C), weakness, abdominal pain, constipation, and anorexia; in severe cases, intestinal perforation occurs. The disease is definitively diagnosed by the **Widal agglutination test**. **Answer: (B)**

Q2. In the life cycle of the malarial parasite *Plasmodium*, the definitive events of sexual reproduction (formation and fusion of gametes) and formation of motile sporozoites occur within the:

(A) Hepatic parenchymal cells of the human host
(B) Erythrocytes of the human host following trophozoite development
(C) Gut lumen and salivary glands of the female *Anopheles* mosquito
(D) Blood plasma of the human host during high fever paroxysms

Sol. Q2. *Plasmodium* exhibits a digenetic life cycle. Asexual multiplication (schizogony) occurs in human liver cells and red blood cells. The sexual stages (gametocytes) initiate in human RBCs but are taken up by the **female Anopheles mosquito**, in whose gut lumen fertilization and zygote (ookinete) formation take place. Subsequent sporogony on the stomach wall produces mature infectious **sporozoites**, which migrate to the mosquito's salivary glands. **Answer: (C)**

Q3. Which of the following cellular and biochemical components of the human body are correctly matched with their designated category of innate immunity barrier?

(A) Lysozyme in tears and saliva — Physical barrier
(B) Mucus coating of the respiratory tract — Physiological barrier
(C) Polymorphonuclear leukocytes (PMNL) and monocytes — Cellular barrier
(D) Interferons secreted by virus-infected cells — Physiological barrier

Sol. Q3. Innate immunity barriers are precisely categorized as: (1) **Physical barriers:** Skin and mucus coating of the respiratory, gastrointestinal, and urogenital tracts; (2) **Physiological barriers:** Gastric acid (HCl), saliva, and lysozyme in tears; (3) **Cellular barriers:** PMNL-neutrophils, monocytes, macrophages, and Natural Killer cells; (4) **Cytokine barriers:** Interferons produced by virus-infected cells. Thus, PMNL and monocytes represent cellular barriers. **Answer: (C)**

Q4. A woman delivers a newborn and breastfeeds the infant during the initial two to three days postpartum. The yellowish fluid secreted by the mammary glands is rich in colostrum. What specific biological protection does this impart to the infant?

(A) Active natural immunity mediated by memory T-lymphocytes
(B) Passive natural immunity conferred by pre-formed dimeric IgA antibodies
(C) Active artificial immunity induced by maternal antigen fragments
(D) Cell-mediated immunity eliminating pathogens through graft rejection

Sol. Q4. Colostrum, the yellowish fluid secreted by mammary glands during the initial days of lactation, contains high concentrations of pre-formed **IgA antibodies**. Because the infant receives ready-made protective immunoglobulins directly without the infant's own immune system undergoing antigenic challenge or generating memory cells, this constitutes a classic example of **natural passive immunity**. **Answer: (B)**

Q5. Human Immunodeficiency Virus (HIV) selectively establishes infection, replicates, and progressively eliminates which vital immunological cell type, ultimately precipitating opportunistic infections?

(A) Cytotoxic CD8+ T-lymphocytes
(B) CD4+ Helper T-lymphocytes (T_H cells)
(C) B-lymphocytes differentiating into plasma cells
(D) Polymorphonuclear neutrophils in the peripheral blood

Sol. Q5. HIV targets cells expressing the CD4 surface receptor, notably **CD4+ Helper T-lymphocytes (T_H cells)** and macrophages. Following viral entry, reverse transcription, and replication, mature virions bud off and lyse the host helper T cells. The progressive decline in CD4+ T-cell count paralyzes both humoral and cell-mediated arms of adaptive immunity, leaving the patient vulnerable to opportunistic pathogens like *Mycobacterium*, *Pneumocystis*, and toxoplasma. **Answer: (B)**

Practice Questions — Part 2 (Q6 to Q10)

Q6. A farmer working in an agricultural field suffers a venomous pit viper snakebite. The attending physician immediately administers an intravenous injection of anti-snake venom. The physiological protection established by this treatment is categorized as:

(A) Natural active immunity
(B) Artificial active immunity
(C) Artificial passive immunity
(D) Innate cellular immunity

Sol. Q6. Anti-snake venom contains pre-formed, concentrated neutralizing antibodies raised in another animal (such as horses or sheep). When injected into a patient facing a life-threatening toxin, these ready-made antibodies provide immediate neutralization without waiting for the host's slow active immune response. Because the antibodies are prepared externally and administered therapeutically, it is classified as **artificial passive immunity**. **Answer: (C)**

Q7. Which of the following biological characteristics fundamentally distinguishes malignant neoplasms (cancers) from benign tumors?

(A) Benign tumors exhibit rapid, invasive infiltration into adjacent muscular tissues
(B) Cancer cells retain strict contact inhibition, growing only in organized monolayers
(C) Malignant tumor cells display the property of metastasis, establishing secondary colonies
(D) Benign tumors shed cells that travel through the bloodstream to lodge in distant vital organs

Sol. Q7. Benign tumors remain strictly localized within a fibrous capsule at their site of origin and do not spread to other anatomical parts. In contrast, malignant neoplasms (cancers) are unencapsulated, grow invasively, and demonstrate **metastasis**—wherein neoplastic cells detach, travel through blood and lymphatic vessels, and colonize distant organs to initiate secondary metastatic tumors. **Answer: (C)**

Q8. Cocaine is an alkaloid that exerts intense central nervous system stimulation and euphoria. Which of the following statements regarding its botanical origin and neurochemical action is correct?

(A) It is extracted from *Papaver somniferum* and binds cannabinoid receptors
(B) It is obtained from *Erythroxylum coca* and interferes with the transport of dopamine
(C) It is derived from *Cannabis sativa* and acts as a central nervous system depressant
(D) It is extracted from *Atropa belladonna* and binds peripheral opioid receptors

Sol. Q8. Cocaine (commonly known as "coke" or "crack") is an alkaloid obtained from the leaves of the South American plant ***Erythroxylum coca***. It acts as a powerful central nervous system stimulant by blocking the reuptake transporter of the neurotransmitter **dopamine** in the brain, resulting in elevated dopamine concentrations that produce euphoria and increased energy. In excessive doses, it triggers acute hallucinations. **Answer: (B)**

Q9. During human gestation, which immunoglobulin class uniquely traverses the placental barrier from the maternal circulation into fetal blood, providing essential humoral defense to the neonate?

(A) Immunoglobulin G (IgG)
(B) Immunoglobulin A (IgA)
(C) Immunoglobulin M (IgM)
(D) Immunoglobulin E (IgE)

Sol. Q9. **Immunoglobulin G (IgG)** is the most abundant circulating antibody class in human serum (80%). Due to its monomeric structure and specialized Fc receptors (FcRn) expressed on trophoblastic cells of the placenta, IgG is the **only immunoglobulin class** capable of crossing the human placenta from mother to fetus, conferring vital passive immunity during late gestation and early infancy. **Answer: (A)**

Q10. Rheumatoid arthritis is a debilitating chronic systemic disease in which synovial joints become inflamed and damaged. In immunological terms, this pathology is classified as:

(A) An infectious protozoan disease transmitted by mosquito vectors
(B) An immediate IgE-mediated allergic hypersensitivity reaction
(C) An auto-immune disorder caused by loss of immunological self-tolerance
(D) A severe combined primary immunodeficiency disease

Sol. Q10. **Rheumatoid arthritis** is an **auto-immune disease**. Under normal physiological conditions, the immune system distinguishes self from non-self antigens. In autoimmune conditions, immunological self-tolerance breaks down, prompting the immune system to produce auto-antibodies (such as rheumatoid factor) that attack and destroy the host's own synovial membranes and cartilage lining the joints. **Answer: (C)**

Practice Questions — Part 3 (Q11 to Q15)

Q11. The infectious developmental stage of *Plasmodium* that is inoculated into human peripheral circulation during the bite of an infected female *Anopheles* mosquito is the:

(A) Trophozoite stage
(B) Sporozoite stage
(C) Merozoite stage
(D) Gametocyte stage

Sol. Q11. The infectious stage of *Plasmodium* for humans is the motile, spindle-shaped **sporozoite**. Sporozoites reside in large numbers inside the salivary glands of an infected female *Anopheles* mosquito and are injected into human subcutaneous capillary vessels during a blood meal. They swiftly migrate to hepatocytes to initiate exo-erythrocytic schizogony. **Answer: (B)**

Q12. Elephantiasis (lymphatic filariasis) is characterized by chronic, disfiguring inflammatory enlargement of the lower limbs and external genitalia. The causative helminth parasite and its biological vector are respectively:

(A) *Ascaris lumbricoides* and the common housefly (*Musca domestica*)
(B) *Wuchereria bancrofti* and the female *Culex* mosquito
(C) *Entamoeba histolytica* and the female *Anopheles* mosquito
(D) *Trichophyton rubrum* and the female *Aedes* mosquito

Sol. Q12. Lymphatic filariasis is caused by filarial nematodes, primarily ***Wuchereria bancrofti*** and *W. malayi*. The parasites are transmitted to humans through the bite of an infected **female Culex mosquito**. Adult filarial worms lodge inside the lymphatic vessels of the lower limbs and scrotum for years, causing progressive chronic inflammation, lymphatic obstruction, and massive swelling (**elephantiasis**). **Answer: (B)**

Q13. Heroin, chemically designated as diacetylmorphine (commonly known on the street as "smack"), is a crystalline depressant produced commercially by:

(A) Demethylation of cocaine extracted from *Erythroxylum coca*
(B) Acetylation of morphine extracted from the latex of *Papaver somniferum*
(C) Oxidation of natural cannabinoids extracted from *Cannabis sativa*
(D) Fermentation of ergot fungus alkaloids growing on rye plants

Sol. Q13. Heroin (diacetylmorphine) is a white, odorless, bitter crystalline compound synthesized by the chemical **acetylation of morphine**. Morphine is a natural phenanthrene alkaloid extracted from the milky latex of the unripened capsules of the opium poppy ***Papaver somniferum***. Heroin binds opioid receptors in the brain and gut, acting as a potent central nervous system depressant that slows physiological functions. **Answer: (B)**

Q14. In clinical oncology, transformed neoplastic cells often evade host immune surveillance. To counteract this, patients are frequently administered biological response modifiers that activate cytotoxic immunity. A primary clinical agent used for this purpose is:

(A) Morphine sulphate
(B) α-Interferon (α-IFN)
(C) Antihistamines
(D) Cyclosporin A

Sol. Q14. Tumor cells frequently develop immunoevasion strategies, downregulating MHC molecules or concealing tumor antigens. In modern immunotherapy, patients are administered **biological response modifiers** such as **α-interferon**, which stimulates host cellular immunity (activating Natural Killer cells, macrophages, and cytotoxic T-lymphocytes) to recognize and destroy neoplastic cells. **Answer: (B)**

Q15. Which of the following pairs of anatomical organs are strictly classified as Primary Lymphoid Organs in the human body?

(A) Spleen and Peyer's patches
(B) Lymph nodes and Tonsils
(C) Bone marrow and Thymus
(D) Spleen and Mucosa-Associated Lymphoid Tissue (MALT)

Sol. Q15. **Primary lymphoid organs** are the anatomical tissues where immature lymphoid stem cells originate, differentiate, and mature into immunocompetent, antigen-sensitive lymphocytes. In humans, the primary lymphoid organs are strictly the **bone marrow** (where all lymphocytes originate and B-cells mature) and the **thymus** (where T-lymphocytes undergo maturation and education). Spleen, lymph nodes, tonsils, and MALT are secondary lymphoid organs. **Answer: (C)**

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